

Chapter III

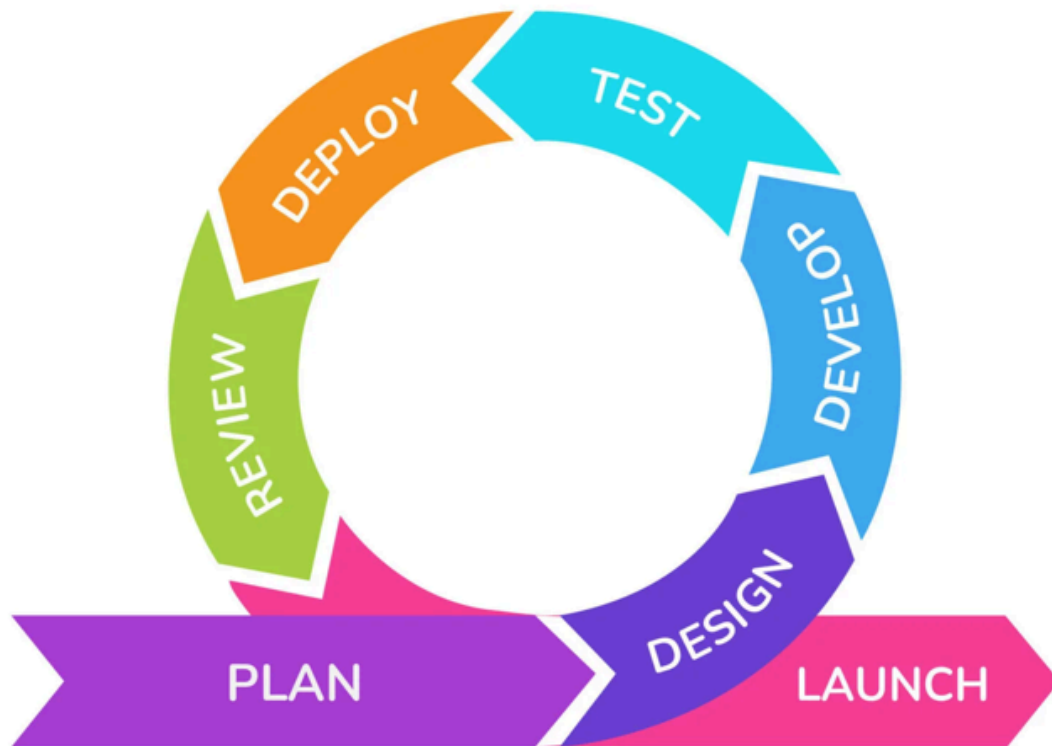
Methodology

This chapter introduces the systematic approach and tools used to develop and evaluate the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration. The study adopts the Systems Development Life Cycle (SDLC) framework, which provides a structured process for planning, analyzing, designing, building, testing, and implementing the system. The chapter outlines the operational framework that guided the project and presents key modeling tools including the Context Diagram and Data Flow Diagrams (DFD) to illustrate system processes, the Entity-Relationship Diagram (ERD) to define data structures, and the Unified Modeling Language (UML) Use Case Diagram and scenarios to describe user interactions. A complete Data Dictionary was provided to define all data elements used in the system. The chapter also specified the necessary hardware and software requirements, presented the project timeline, and discussed the evaluation process, which includes system testing and assessment using the ISO/IEC 25010:2011 Software Quality Model to ensure functionality, reliability, and overall user experience.

Systems Development Method

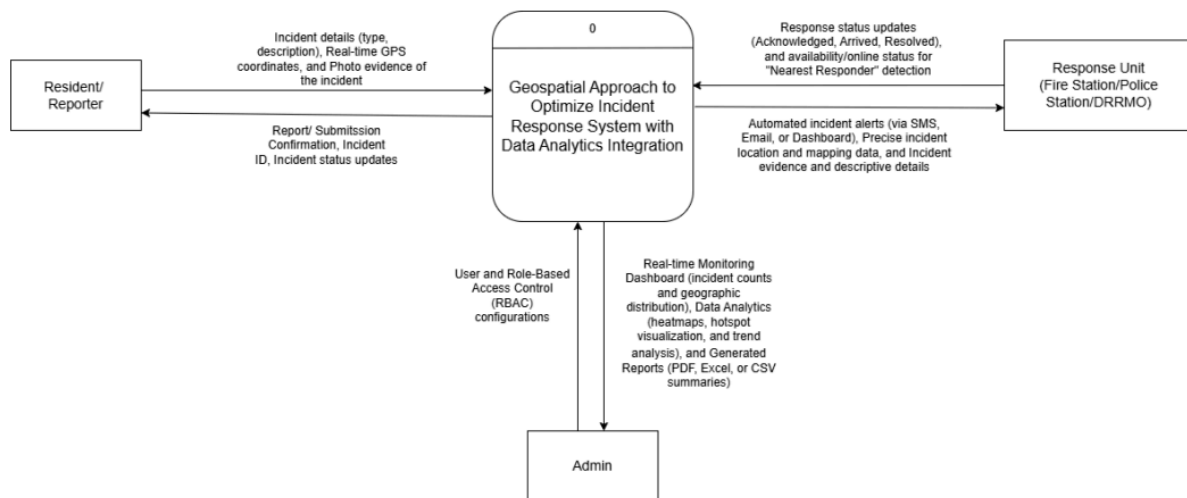
The systems development method refers to the framework used to structure, plan, and control the process of developing the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration. The proponents used the Agile methodology, specifically an incremental development model, which allows flexibility and continuous feedback throughout the system development process.

Agile development method of Geospatial Approach to Optimize Incident Response System with Data Analytics Integration.



Data Flow Diagram

Context Diagram of Geospatial Approach to Optimize Incident Response System with Data Analytics Integration



The context diagram of the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration, illustrating how data from the Resident/Reporter, Response Units, and the Administrator is collected and processed to generate incident reports, automated alerts, and analytics outputs for improved emergency response delivery.

The Level 1 data flow of the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration. This system manages information and coordinates emergency response for human-induced incidents affecting residents within the jurisdiction of the Talisay City DRRMO, BFP Station, and Police Station.

Incident Logs and Records data store. The data is then passed to the Incident Type Classification Module, which categorizes the incident by type and assigns a severity level, storing the results in the Classification data store.

The Data Validation and Logging Module verifies the completeness and accuracy of the submitted incident report details, including photo evidence, and logs confirmed records into the Response Logs data store. The Geofencing Area Detection Module then extracts GPS coordinates, identifies the barangay or zone jurisdiction, allocates the matched response unit, and produces an area mapping report delivered to the assigned Response Unit.

The Alert and Response Coordination Module detects the nearest available response unit, composes the alert notification, and dispatches it to the appropriate agency DRRMO, BFP, or Police Station through the system dashboard, SMS, or email. The Incident Monitoring and Status Updates Module receives response status updates from the Response Unit and notifies the Resident/Reporter of the current incident status, logging all status changes to the User/Resident Accounts Records data store.

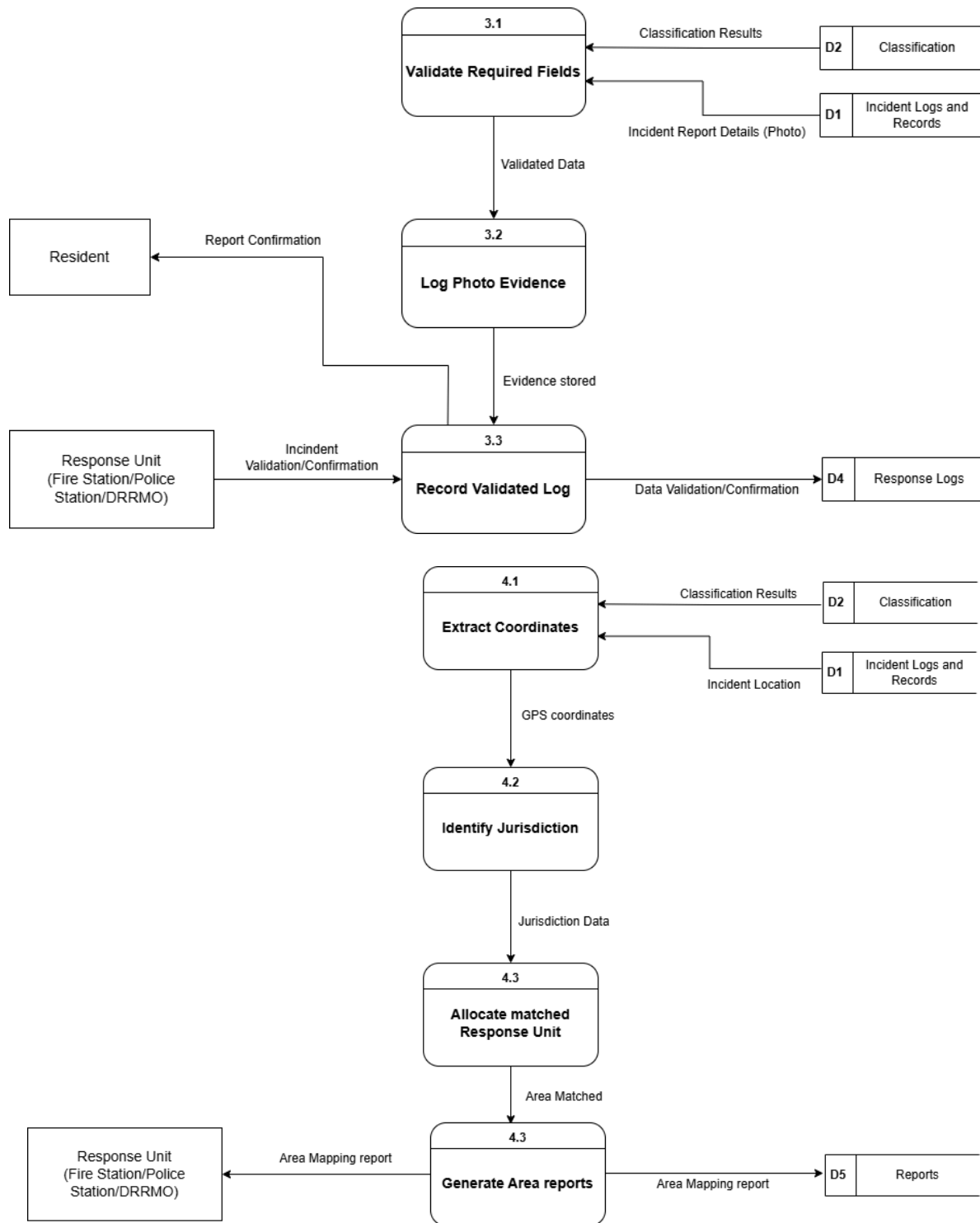
The Hotspot and Trend Analysis Module aggregates incident data, computes heatmaps, identifies trend patterns, and delivers mapping visualizations and trend reports to the Response Unit. Finally, the Generate Reports Module compiles all recorded data, formats and generates structured reports, and exports incident summaries in PDF, Excel, or CSV format for administrative and planning use.

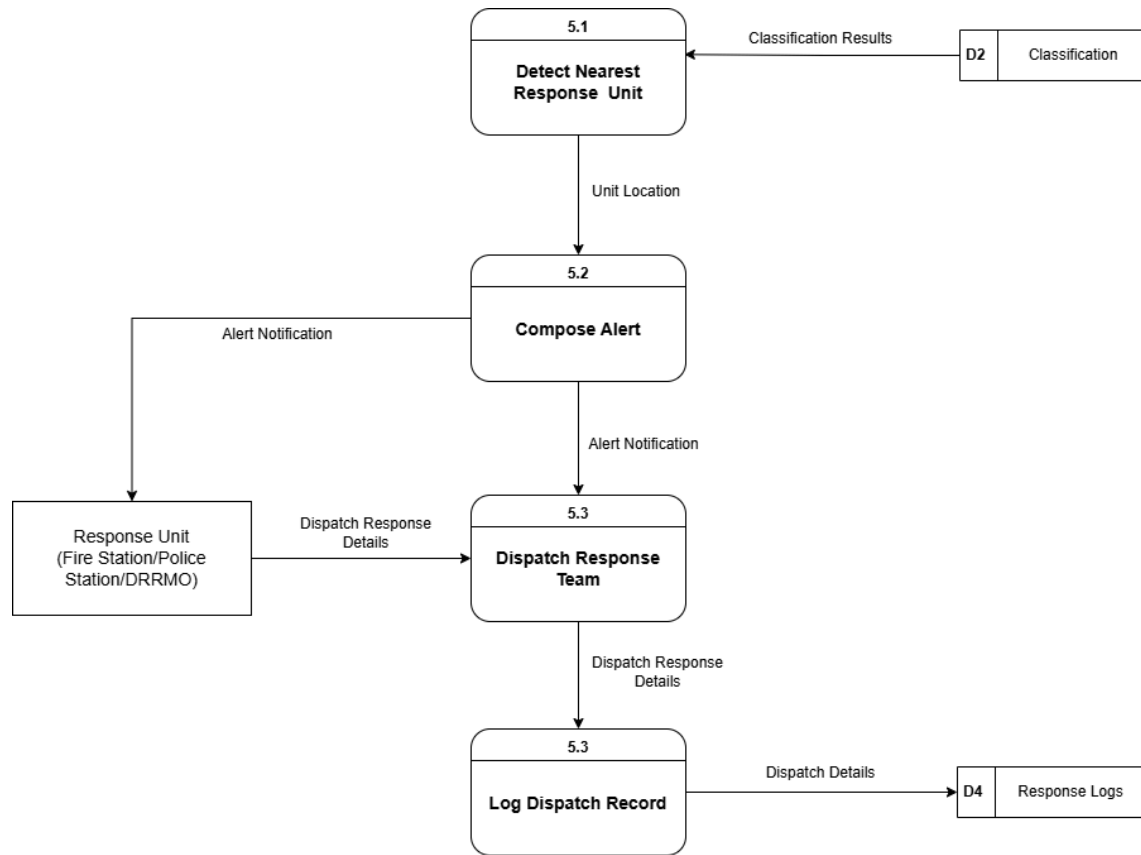
In summary, this system connects all critical incident management processes in one centralized platform, enabling the Talisay City DRRMO, BFP Station, and Police Station to detect, coordinate, monitor, and analyze human-induced incidents more efficiently and effectively.

Figure 5

Level 2 Detailed Data Flow Diagram for Incident Reporting Process of the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration







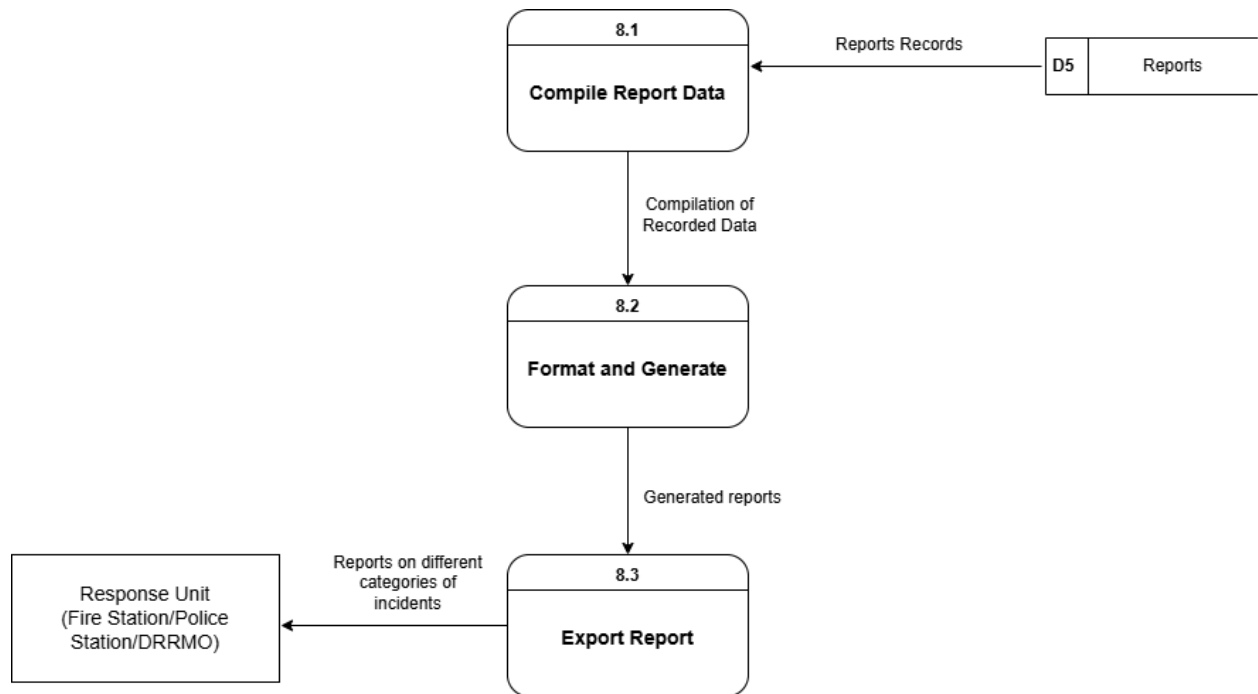
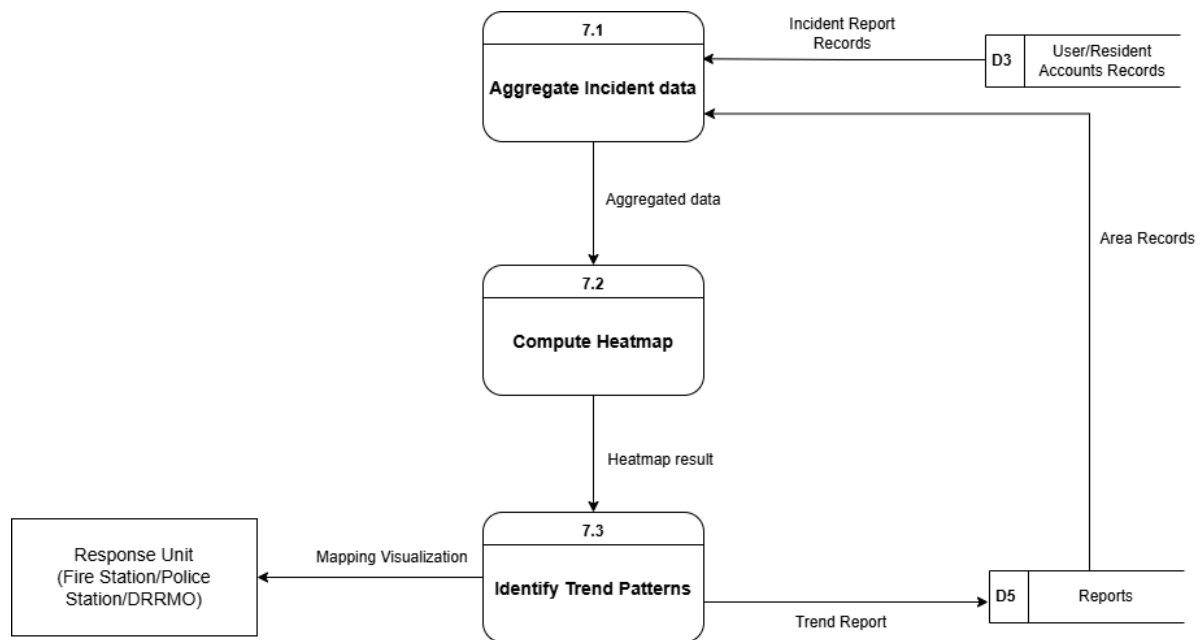


Figure 5 illustrates the detailed data flow of the Incident Report Recording and Management process within the Geospatial Approach to Optimize Incident Response System, focusing on the end-to-end handling of incident submissions from residents to the point of response unit notification.

The process begins with the **Receive Incident Data** subprocess, which collects raw incident details submitted by the Resident/Reporter, including incident type, location coordinates, and photo evidence. This data is then passed to the **Validate Report Fields** subprocess, which checks for the completeness and correctness of all required inputs.

Following validation, the **Generate Incident ID** subprocess creates a unique incident code and links the record to the User/Resident Accounts data store to associate the report with the submitting user. The validated and identified record is then handled by the **Store Incident Record** subprocess, which saves the incident to the Incident Logs and Records data store and simultaneously sends the incident report details to the assigned Response Unit, triggering the response coordination workflow.

In essence, this detailed process centralizes and organizes all critical incident data, enabling the Talisay City LGU response agencies to act on verified, accurately recorded incidents and reduce the time between reporting and on-site response.

Entity Relationship Diagram

Figure 6

Entity Relationship Diagram of the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration

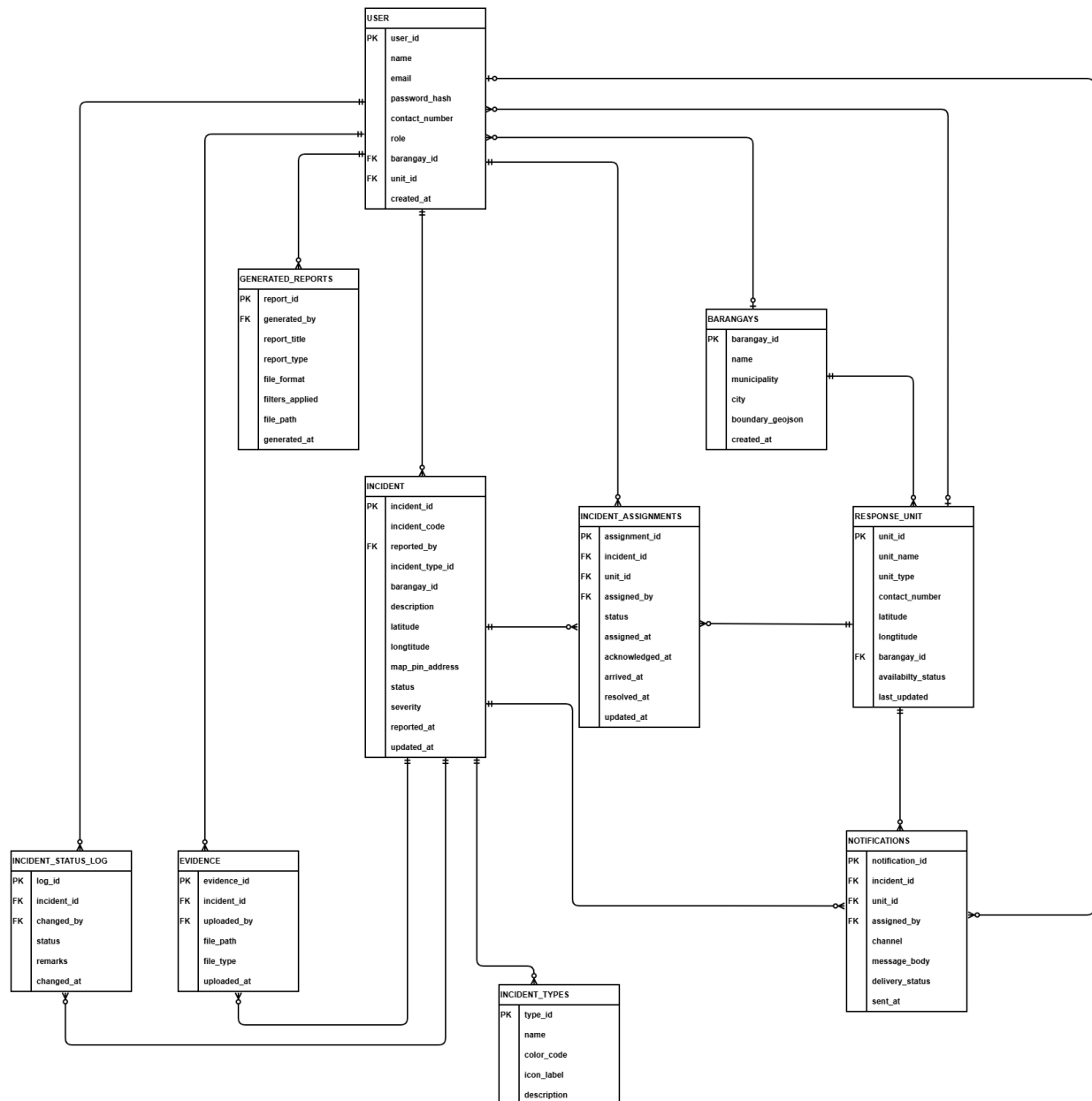


Figure 6 shows the Entity Relationship Diagram (ERD) that illustrates the structure and relationships of the database tables within the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration. It depicts the connections among the system's core entities — USER, BARANGAYS, RESPONSE_UNIT, INCIDENT, INCIDENT_TYPES, INCIDENT_ASSIGNMENTS, NOTIFICATIONS, EVIDENCE, INCIDENT_STATUS_LOG, and GENERATED_REPORTS — and their attributes, providing a comprehensive overview of how incident data, geospatial records, user accounts, and response logs are organized to support the system's functionality and data management requirements.

Use Case Diagram

Figure 7

Use Case Diagram of the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration

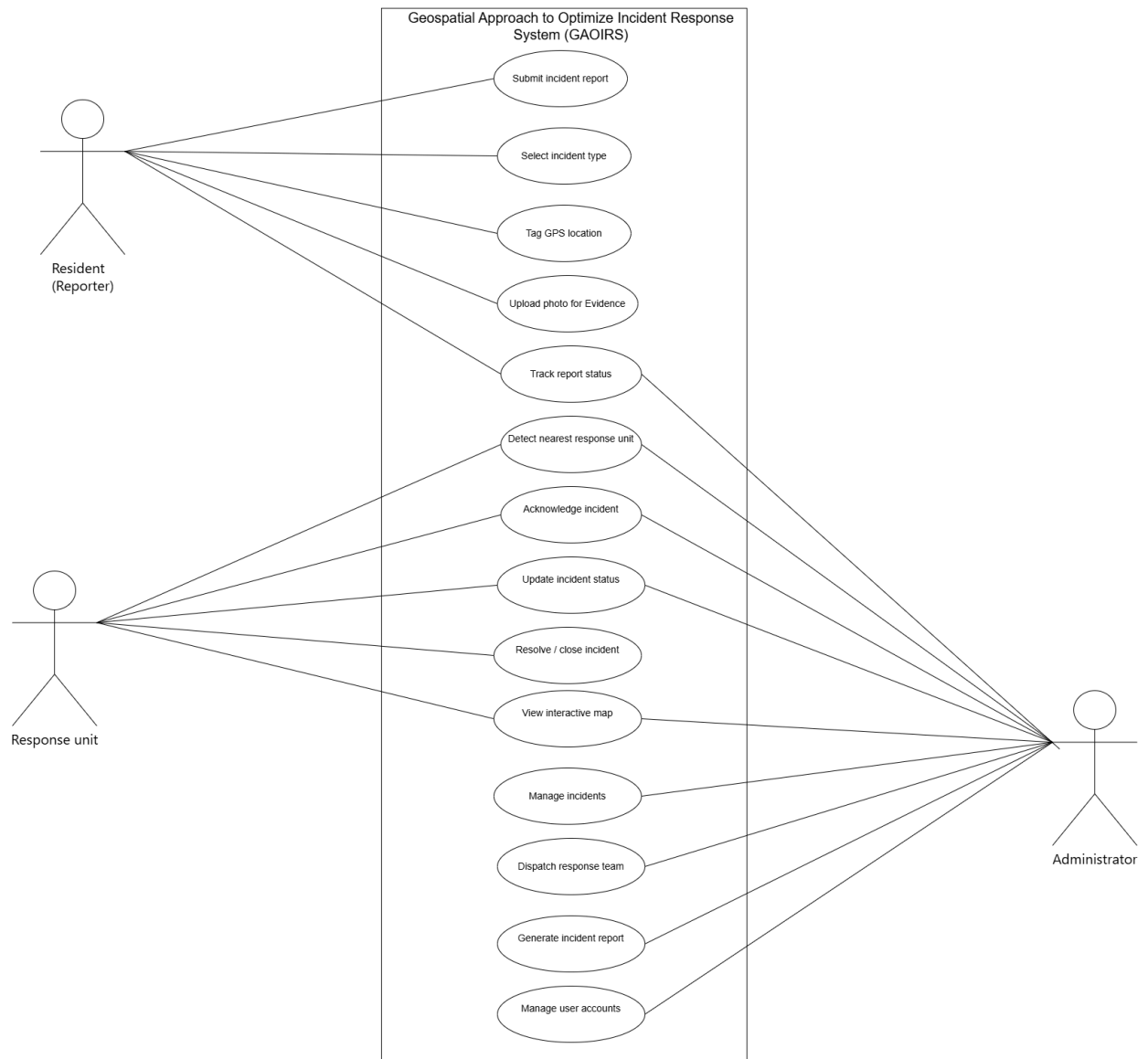


Figure 7 shows the use case diagram of the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration. It explains the main functions of the system and the

actors who interact with it: the Resident/Reporter, the Response Unit (DRRMO, BFP, Police Station), and the Administrator.

The **Resident/Reporter** can submit incident reports with GPS coordinates and photo evidence, receive submission confirmations, and track incident status updates through the system.

Response Unit — updated to match the text: now includes receiving automated alerts (dashboard/SMS/email), viewing incident location & map data, acknowledging, updating response status (Acknowledged → Arrived → Resolved), resolving/closing, and accessing hotspot & trend analysis.

Administrator — expanded to reflect all described capabilities: managing user accounts & access control, monitoring the real-time dashboard, viewing heatmap & hotspot analysis, generating & exporting reports (PDF/Excel/CSV), and overseeing the full incident lifecycle.

DATA DICTIONARY

Users

Field	Field Type	Length	Key	Description
user_id	VARCHAR	20	Primary Key (PK)	Unique identifier for each user
Name	VARCHAR	100		Full name of the user
Email	VARCHAR	100		User email address
password_hash	VARCHAR	255		Encrypted password
contact_number	VARCHAR	13		User contact number
role	ENUM	-		User role (admin, responder, citizen)
barangay_id	INT	-	Foreign Key (FK)	Linked barangay
unit_id	INT	-	Foreign Key (FK)	Linked response unit
created_at	DATETIME	-		Account creation date

BARANGAYS Table

Field	Field Type	Length	Key	Description
barangay_id	INT	-	Primary Key (PK)	Unique barangay ID
name	VARCHAR	100		Barangay name
municipality	VARCHAR	100		Municipality name
city	VARCHAR	100		City name
boundary_geojson	TEXT	-		Geographic boundary data
created_at	DATETIME	-		Date created

RESPONSE_UNIT Table

Field	Field Type	Length	Key	Description
unit_id	INT	-	Primary Key (PK)	Unique barangay ID
unit_name	VARCHAR	100		Name of response unit
unit_type	VARCHAR	50		Type (police, fire, medical)
contact_number	VARCHAR	20		Contact number
latitude	DECIMAL	-		Location latitude
longitude	DECIMAL	-		Location longitude
barangay_id	INT	-	Foreign Key (FK)	Assigned barangay
availability_status	ENUM	-		Availability status
last_updated	DATETIME	-		Last updated

INCIDENT Table

Field	Field Type	Length	Key	Description
incident_id	INT	-	Primary Key (PK)	Unique barangay ID
incident_code	VARCHAR	50		Unique incident code

reported_by	INT	-	Foreign Key (FK)	User who reported
incident_type_id	INT	-	Foreign Key (FK)	Incident type
barangay_id	INT	-	Foreign Key (FK)	Location barangay
description	TEXT	-		Incident details
latitude	DECIMAL	-	Foreign Key (FK)	Latitude
longitude	DECIMAL	-		Longitude
map_pin_address	VARCHAR	255		Address of incident
status	ENUM	-		Incident status
severity	ENUM	-		Severity level
reported_at	DATETIME	-		Date reported
updated_at	DATETIME	-		Last update

INCIDENT_TYPES Table

Field	Field Type	Length	Key	Description
type_id	INT	-	Primary Key (PK)	Unique barangay ID
name	VARCHAR	100		Barangay name
color_code	VARCHAR	20		Display color
icon_label	VARCHAR	50		Icon label
description	TEXT	-		Description

INCIDENT_ASSIGNMENTS Table

Field	Field Type	Length	Key	Description
assignment_id	INT	-	Primary Key (PK)	Unique assignment ID
incident_id	INT	-	Foreign Key (FK)	Linked incident
unit_id	INT	-	Foreign Key (FK)	Assigned unit
assigned_by	INT	-	Foreign Key (FK)	User who assigned
status	ENUM	-		Assignment status

assigned_at	DATETIME	-		Date assigned
acknowledged_at	DATETIME	-		Acknowledged time
arrived_at	DATETIME	-		Arrival time
resolved_at	DATETIME	-		Resolution time
updated_at	DATETIME	-		Last updated

NOTIFICATIONS Table

Field	Field Type	Length	Key	Description
notification_id	INT	-	Primary Key (PK)	Unique assignment ID
incident_id	INT	-	Foreign Key (FK)	Related incident
unit_id	INT	-	Foreign Key (FK)	Target unit
assigned_by	INT	-	Foreign Key (FK)	Sender
channel	VARCHAR	50		Notification type
message_body	TEXT	-		Message content
delivery_status	ENUM	-		Delivery status
sent_at	DATETIME	-		Date sent

EVIDENCE Table

Field	Field Type	Length	Key	Description
evidence_id	INT	-	Primary Key (PK)	Unique assignment ID
incident_id	INT	-	Foreign Key (FK)	Related incident
uploaded_by	INT	-	Foreign Key (FK)	User uploader
file_path	VARCHAR	255		File location
file_type	VARCHAR	50		File type

uploaded_at	DATETIME	-		Upload date
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INCIDENT_STATUS_LOG Table

Field	Field Type	Length	Key	Description
log_id	INT	-	Primary Key (PK)	Unique log ID
incident_id	INT	-	Foreign Key (FK)	Related incident
changed_by	INT	-	Foreign Key (FK)	User who changed
status	ENUM	-		Updated status
remarks	TEXT	50		Notes
changed_at	DATETIME	-		Date changed

GENERATED_REPORTS Table

Field	Field Type	Length	Key	Description
report_id	INT	-	Primary Key (PK)	Unique log ID
generated_by	INT	-	Foreign Key (FK)	User who generated
report_title	VARCHAR	150		Report title
report_type	VARCHAR	50		Report type
file_format	VARCHAR	20		File format
filters_applied	TEXT	-		Applied filters
file_path	VARCHAR	255		File location
generated_at	DATETIME	-		Date generated

Hardware and Software Requirements

The Hardware and Software Requirements provide the detailed minimum specifications needed to support the functions and execution of the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration.

Software Requirements

- Operating System: Windows 11, macOS
- Web Server: Apache or Nginx, with hosting services such as Hostinger or GoDaddy for web hosting
- Database: PostgreSQL for relational data management
- Programming Languages:
 - Frontend: HTML, CSS, JavaScript (with Bootstrap 5 and Leaflet.js)
 - Backend: Node.js
- Frameworks and Libraries: Express.js for server-side development; Chart.js for data visualization
- GIS Software: Leaflet.js for interactive geospatial mapping, geofencing, and nearest responder detection
- Version Control: Git for source code management
- API Integration: RESTful APIs for data exchange and external notification services

Recommended Minimum Hardware Requirements

The following is the list of hardware devices needed to support the functions and execution of the proposed system:

- Computer: A desktop or laptop computer with internet access to host and access the system
- Processor: A multi-core processor with a speed of at least 2.0 GHz
- RAM: At least 8 GB of RAM for smooth system operation and multitasking
- Storage: Minimum of 100 GB of free storage space to accommodate system files, database records, uploaded evidence, and generated reports
- Network: Stable internet connection with at least 10 Mbps download speed for optimal real-time performance, geospatial map rendering, and automated notification delivery
- GPS-capable devices (smartphones or tablets): Required for accurate GPS coordinate capture during mobile incident reporting by field reporters or community members

